

Name:

Date:

Period:

Equation Sheet - Spring Final Exam

Momentum

$$F \cdot t = \Delta p$$

$$\Delta p = p_f - p_i$$

$$\Sigma p_i = \Sigma p_f$$

$$p = mv$$

Energy

$$W = Fd$$

$$F_G = mg$$

$$P = \frac{W}{t}$$

$$KE = \frac{1}{2}mv^2$$

$$PE = mgh$$

$$KE_i + PE_i + W = KE_f + PE_f$$

Simple Harmonic Motion

$$T_P = 2\pi\sqrt{\frac{L}{g}}$$

$$T_S = 2\pi\sqrt{\frac{m}{k}}$$

$$F_S = -kd$$

$$F_G = mg$$

$$v = f\lambda$$

$$f = \frac{1}{T}$$

Light & Sound

$$v = f\lambda$$

$$f = f_s \left(\frac{v \pm v_o}{v \mp v_s} \right)$$

$$\frac{1}{d_i} + \frac{1}{d_o} = \frac{1}{f}$$

$$M = \frac{h_i}{h_o} = \frac{-d_i}{d_o}$$

Speeds of Sound:

air: 340 m/s

water: 1530 m/s

iron: 5100 m/s

Speed of Light: 3.0×10^8 m/s